

Welcome to

Network Attacks I – Concepts

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Slides are available as PDF, https://codeberg.org/kramse
academy-2026-day1-network-attacks-I.tex in the repo security-courses

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Goals for today



- Explain common network-attack concepts: botnets and DoS/DDoS
- Explain amplification and reflection attacks
- Explain SYN, UDP and ICMP flooding
- Explain ARP poisoning and man-in-the-middle risks

Plan for today



- Attack overview and the DDoS problem
- Flooding and availability attacks
- TCP/IP protocol weaknesses
- ARP poisoning and man-in-the-middle
- DNS and SNMP problems

Introduction: Attack overview



Source: http://www.sicherheitstacho.eu/

The Internet and DDoS is trouble



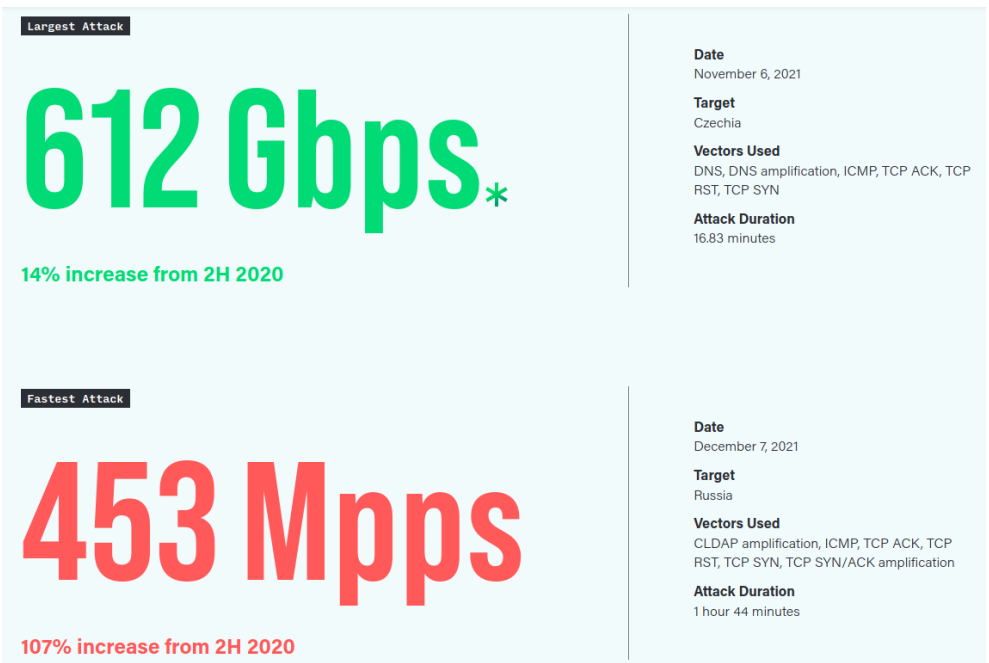
Security attacks and DDoS is very much in the media



Source: Netscout Global DDoS Threat Intelligence Report 2nd half 2021

https://www.netscout.com/threatreport/global-ddos-attack-trends/

DDoS Attacks are HUGE



Extremely hard to protect against from a small network

Source: https://www.netscout.com/threatreport/global-ddos-attack-trends/

Availability and Network flooding attacks



The attacks we are discussing today are:

- **SYN flood** is the most basic and very common on the internet towards 80/tcp and 443/tcp
- **ICMP and UDP flooding** are the next popular targets – more similar ones exist
- Special packets and protocols – anything that can create *load on systems* work
- All of them try to use up some resources
- **Memory space** in specific sections of the **kernel, TCP state, firewalls state, number of concurrent sessions/connections**
- **Interrupt processing** of packets - packets per second (pps)
- **CPU processing** in firewalls, pps
- CPU processing in server software
- **Bandwidth** - megabits per second (mbps)
- Typically source is spoofed or amplification attacks abusing devices on the Internet

Botnets



- **FIXME:** add botnet concept + an accurate example (e.g. Mirai) – not in these decks

SYN / UDP / ICMP flooding



- **FIXME:** per-flood concept slide (these are shown hands-on in part II); subject plan obj 9

Security problems in the TCP/IP Suite



The title of a nice paper, and the rest of today

> The paper "Security Problems in the TCP/IP Protocol Suite" was originally pub- lished in Computer Communication Review, Vol. 19, No. 2, in April, 1989

Problems described in the original:

- sequence number spoofing
- routing attacks,
- source address spoofing
- authentication attacks

TCP sequence number prediction



> TCP SEQUENCE NUMBER PREDICTION One of the more fascinating security holes was first described by Morris [7] . Briefly, he used TCP sequence number prediction to construct a TCP packet sequence without ever receiving any responses from the server. This allowed him to spoof a trusted host on a local network.

tidligere baserede man login/adgange på source IP adresser, address based authentication

Er ikke en pålidelig autentifikationsmekanisme

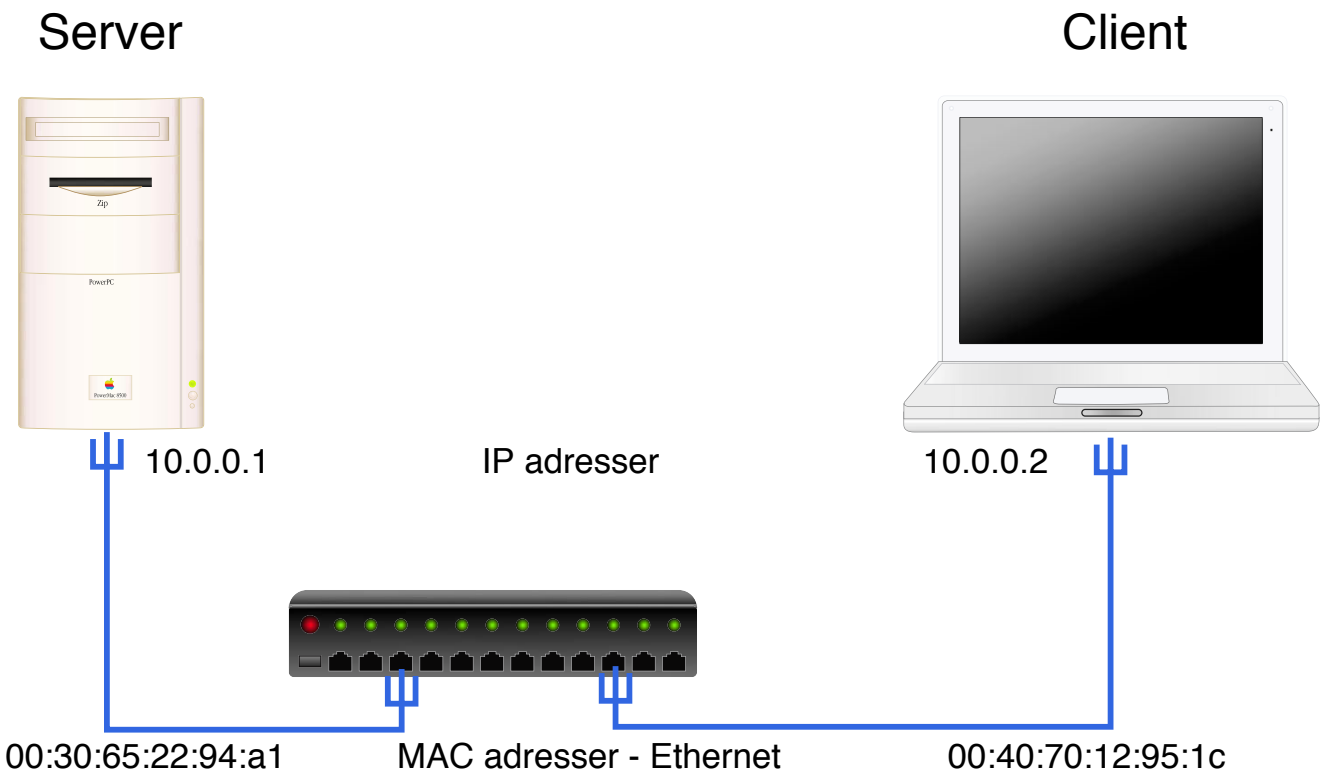
Mest kendt er nok Shimomura der blev hacket på den måde,
måske af Kevin D Mitnick eller en kompagnon

I praksis vil det være svært at udføre på moderne operativsystemer

Se evt. http://www.takedown.com/ (filmen er ikke så god ;-))

Det er naturligvis fint med filtre så man kun kan tilgå services FRA bestemte IP

Address Resolution Protocol (ARP)



Person in the middle attacks



ARP spoofing, ICMP redirects, the classics

Used to be called Man in The Middle MiTM

- ICMP redirect
- ARP spoofing
- Wireless listening and spoofing higher levels like airpwn-ng https://github.com/ICSec/airpwn-ng

Usually aimed at unencrypted protocols

Today we only talk about getting the data, not how to perform higher level attacks

DNS problems



> The Domain Name System (DNS) [32][33] provides for a distributed database mapping host names to IP addresses. An intruder who interferes with the proper operation of the DNS can mount a variety of attacks, including denial of service and password collection. There are a number of vulnerabilities.

We have a lot of the same problems in DNS today

Plus some more caused by middle-boxes, NAT, DNS size, DNS inspection

- DNS must allow both UDP and TCP port 53
- Your DNS servers must have updated software, see DNS flag day https://dnsflagday.net/ after which kludges will be REMOVED!

SNMP problems



> 5.5 Simple Network Management Protocol The Simple Network Management Protocol (SNMP) [37] has recently been defined to aid in network management. Clearly, access to such a resource must be heavily protected. The RFC states this, but also allows for a null authentication service; this is a bad idea. Even a "read-only" mode is dangerous; it may expose the target host to netstat-type attacks if the particular Management Information Base (MIB) [38] used includes sequence numbers. (T

True, and we will talk more about SNMP later in this course.

Summary



- Network attacks span flooding (availability) and protocol abuse (ARP, TCP/IP, DNS)
- Next: the hands-on half – generating and defending against these attacks